THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATIVE AND LOCAL GOVERNMENT

ARUSHA REGION

FORM ONE TERMINAL EXAMINATION MAY 2025

PHYSICS

TIME; 2:30HRS

INSTRUCTIONS

1. This paper consists of section A and B and one question from section C.
2. Answer all questions
3. When necessary the following constant may be used;
   i. Acceleration due to gravity g= 10N/kg
   ii. Density of water = 1000kg/m^3 or 1g/cm^3
   iii. Pie 3.14

SECTION A (16 MARKS)

1. For each item (i-x) choose the correct answer from among the given alternative and write its letter besides the box provided.
   i) How does physics contribute to our understanding of natural world and the universe?
      (A) Physics is not related to the study of the natural world and the universe
      (B) Physics helps us understand the behavior of matter and energy from subatomic particle to celestial bodies
      (C) Physics only deals with artificial and human made system
      (D) Physics is not essential for understanding the natural world and the universe.
   ii) Physics is the study of matter, energy and interaction existing between them. what is the significance of the interaction of matter and energy in physics?
      (A) It helps us to understand the human body and its functions
      (B) It allows us to explore different planets and their atmospheres
      (C) It helps explain the behavior of object and phenomena in the universe
      (D) It is not relevant to the study of physics
   iii) Which of the following instrument is most suitable for measuring the internal diameter of 100ml beaker?
      (A) Meter ruler
      (B) Vernier caliper
      (C) Measuring cylinder
      (D) External caliper

iv) A cuboid with length 3m, width 4m and height 10cm is made from wood .the actual volume of cuboid is (A) $120cm^3$ (B) $1.2m^3$ (C) $24Omm^3$ (D) $120m^3$

v) A force acting on an object causes some properties of the object to change. which of the list contains properties that can be changed by the action of force?
(A) Mass, motion and shape
(B) Mass, motion, size
(C) Mass ,shape, size
(D) Motion , shape, size

vi) A liquid has a volume of $100cm^3$ and mass of 85g. the density of water is$1g/cm^3$.how does the density of the liquid compare with the density of water.
(A) Its density is higher than that of water
(B) Its density is lower than that of liquid
(C) Its density is the same as that of water
(D) It is impossible to say with only this data.

vii) A scuba diver wants to stay motionless at a specific depth in the water. What should the diver do to achieve this?
(A) Add weight to become more negative buoyant
(B) Inflate the buoyancy control device (BCD) to become more positively buoyant
(C) Do not make any adjustments, the diver will automatically maintain neutral buoyancy
(D) Kick their legs continuously to stay in place.

viii) A student wants to compare the densities of two different liquid. Which instruments should the student use for the direct comparison?
(A) Barometer (B) Thermometer (C) hygrometer (D) Hydrometer

ix) If the relative density of substance is 11.3, the density of that substance will be?
(A) $11.3kg/m^3$ (B) $11.3g/cm^3$ (C) $0.000113g/m^3$ (D) $11300kg/cm^3$

x) According to Archimedes principle an object immersed in a fluid it experiences an upthrust that is equal to the weight of the fluid displaced. Upthrust is most affected by;
(A) Volume of immersed object, density of water and force of gravity
(B) Density of fluid displaced, acceleration due to gravity and mass of an object
(C) Weight of an object, force acting on an object and density of water
(D) Density of fluid in which an object immersed, volume of immersed object and acceleration due to gravity

questions	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answers										

2. Match each of the description of the term used in LIST A with the corresponding concept used in LIST B by writting the letter of the response below the item in the table provided

LIST A	LIST B
I. Viscous force II. Stretching force III. Compressional forces IV. Torsional force V. Frictional forces	A. A force that includes electric and magnetic field B. Force that pull the object toward the earth surfaces C. The forces that opposes motion between the two surfaces of an object in contact D. Forces produced on a solid twisted object

	E. Force exerted on an object when its two ends are pulled apart F. Force that resist the movement of an object through the air G. Force exerted on an object when its two ends are pulled toward each other H. Resistance force provided by a fluid when subjected to a tangential force. I. Force exerted on an object by stretched material.

SECTION B (70 marks)

3. (a) Onyango a form one student was select by the chairman of his village to educate his people at Kimbiji village on how studying physics can bring changes in his community so that can lead to development of modern society, having roads, telephone ,cars, tractors and syringe and x-rays in the hospitals. As a form one students help Onyango by giving five point on how physics can contribute to the development of modern society.
   (b) A science class is learning about the properties of material and observes that some objects sink in water while others float even though they are made of the same material.
   (i) state the name of the principle related to the above scenarios
   (ii) state the principle named above
4. (a) MR.Owino is an expert on the field of physics, he wants to measure the diameter of a ball bearing in the laboratory. As a young scientist help him to choose a proper instrument for measuring that ball bearing.
   (i) State the name of the instrument can be used
   (ii) State the accuracy of that instrument used in centimeter
   b. In the process of measuring the diameter of ball bearing he found that the least of screw gauge is 0.01mm, the sleeve shows a reading of 1.6cm and thimble indicate 48 divisions. Calculate the accurate diameter of ball bearing.
5. (a) from the listed objects classify them of regular or irregular shape; cube, orange, stone , banana ,water pipe, house, cuboid, chalkboard.
   (B) The internal and external radius of a hollow silver sphere are 7cm and 14cm respectively.find the volume of the silver sphere.
6. In an experiment to determine the weight of coloured water a student was given a beam balance ,beaker and colour water.if the mass of water was measured to be 620g.calculate the weight of water.
7. (a) explain the following terms in relation to the concept of sinking and floating
   (i) Buoyancy
   (ii) Apparent weight
   (iii) Actual weight
   c. State three condition that satisfied for an object to float in water.
   d. A body immersed in water displaces 1.1N . if its weight while in water was 3.3N. Find its weight in air
8. (A) (I) state two difference between density and relative density
   (ii) density and relative density are different but has many application in our life everyday, state four application of density and relative density.
   (iii) a form one student in physics class obtained the following results from an experiment;
   Mass of empty beaker = 48g

Mass of beaker + liquid = 60g

If she had used the pipette of $20cm^3$ to transfer liquid to beaker.calculate

(i) Density of liquid

(ii) Relative density of liquid in kg/m^3

9. An object is hung in spring balance. If it weight 40N in air and 30N when immersed in water. Calculate; (i) upthrust on the object

(II) weight of the displaced water

(iii) Mass of displaced water

(iv) Volume of displaced water

(v) Relative density of the object

SECTION C(15 marks)

10. (a) A form one student is in physics laboratory. He is provided with density bottle, sand, digital balance and water. He is required to determine the density of sand using the instruments provided above. show how he will determine the density of sand.

(b) in an experiment above he obtained the following results

Mass of density bottle (M_1) =200g

Mass of density bottle and sand (M_2) =490g

Mass of density bottle, sand and water (M_3) =550g

Mass of density bottle and water (M_4) = 300g

(i) What is the density of sand

(ii) Determine the relative density of sand